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SUPERSEDING
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MILITARY SPECIFICATION

SULFUR AND SALICYLIC ACID OINTMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Sulfur and Salicylic Acid Ointment (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

* 3.1 Material. Shall be an ointment consisting of the following:

Sulfur Precipitated, USP	2 percent
Salicylic Acid, USP	2 percent
Ointment Base 1/	96 percent

1/ The ointment base shall be a penetrating type with surface active combinations of soapless cleansers and wetting agents, including sodium dioctyl sulfosuccinate, sodium alkyl aryl polyether sulfonate, and sodium lauryl sulfoacetate.

3.2 Assay of finished ointment.

* 3.2.1 Sulfur. Shall assay to contain not less than 1.80 percent and not more than 2.20 percent of sulfur when determined as specified in 4.7.1.

* 3.2.2 Salicylic Acid. Shall assay to contain not less than 1.80 percent and not more than 2.20 percent of sulfur when determined as specified in 4.7.2.

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3.3 pH. The pH of the finished ointment shall not be less than 2.60 and not more than 3.10 when determined as specified in 4.7.3.

3.4 Loss on drying. The loss on drying of the finished ointment shall be not less than 57.5 percent and not more than 59.5 percent, when determined as specified in 4.7.4.

3.5 Lather volume and permanence tests. The total volume of lather produced by the finished ointment shall be not less than 45 ml when read immediately after testing and not less than 35 ml after standing for 30 minutes, when tested as specified in 4.7.5.

3.6 Ingredients.

3.6.1 Salicylic Acid. The salicylic acid used in the manufacture of the ointment shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.6.2 Sulfur. The sulfur used in the manufacture of the ointment shall be Precipitated Sulfur and shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.7 Other ingredients. All other ingredients used in the manufacture of the ointment shall be of USP or NF quality, or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.8 Ten-week separation test. The finished ointment shall not show any gross separation when tested as described in 4.7.6.

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3.9 Delivery. Not more than 4 months shall have elapsed from the date of manufacture of the ointment, to the date of delivery to the Government.

3.10 Workmanship. The material and its containers shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of sulfur and salicylic acid ointment produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

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4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

4.3.1 End item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample in tubes
Assay for sulfur	3.2.1	S-2 To obtain a composite sample to be utilized in performing these tests	5
Assay for salicylic acid	3.2.2		
pH	3.3		
Loss on drying	3.4		
Lather volume and permanence test	3.5		
Separation test	3.7		12

1/ Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished ointment shall be examined to determine compliance with all requirements of this specification. Non-comformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

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TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Ointment not free from foreign matter.
102	Ointment has crystal formation.
103	Ointment not homogeneous; not free from separation.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package. packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Assay for sulfur in the finished ointment.

Standard solution:

Prepare a solution of Sulfur, USP, in benzene having a known final concentration of approximately 0.2 mg per ml.

Procedure:

Accurately weigh approximately 100 mg of ointment into a beaker, disperse in 20 ml of purified water, and transfer quantitatively to a 60 ml separator using small portions of purified water. Pipet 10.0 ml methylene chloride, reagent grade, to the separator; stopper and shake well. Permit the layers to separate, and run off approximately 5 ml of the lower layer (organic solvent) through phase separating paper such as Whatman 1PS, or equivalent. Collect the clear filtrate in an Erlenmeyer flask and stopper to prevent evaporation.

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* Using a prepared thin layer chromatography plate containing silica gel GS, with a uniform layer thickness of approximately 250 microns, separately spot 10 ul of standard solution and 10 ul of clear filtrate sample solution approximately 4.0 cm from the bottom of the plate. Develop to a height of approximately 10 cm using iso-octane and methylene chloride as the mobile solvent in a solvent chamber previously saturated with iso-octane and methylene chloride.

Permit the plats to dry thoroughly in air at room temperature, and quantitate the spots corresponding to sulfur (Rf approximately 0.7) using a recording densitometer, with exciting light at 254 nm and emitted light measured at 520 nm. Set the instrument for optimal operating conditions. Integrate the areas under the curve for standard and sample.

Calculation:

$$\frac{Qu \times CS \times 1000}{Qs \times \text{weight sample (mg)}} = \text{percent sulfur in ointment}$$

where:

Qu = quantitation value sample
QS = quantitation value standard
Cs = concentration standard mg per ml

4.7.2 Assay for salicylic acid in the finished ointment.

Reagent:

Dissolve 1.0 gram $Fe(NO_3)_3$ and 1.0 ml concentrated hydrochloric acid in 100 ml purified water.

* Standard solution:

Accurately weigh approximately 40 mg of salicylic acid reference standard into a 100 ml volumetric flask. dissolve in and dilute to volume with alcohol.

Standard may be obtained from Technical Operations Division, Directorate of Medical Materiel, Defense Personnel Support Center, ATTN: DPSC-ATQ, 2800 South 20th Street, Philadelphia, PA 19101.

* Sample solution:

Weight accurately approximately 200 mg of sample into a tared 100 ml volumetric flask. Add 20 ml alcohol and shake well. Add about 50 ml purified water, 5.0 ml $Fe(NO_3)_3$ reagent and dilute to volume with purified water. Shake and allow to stand for at least 15 minutes.

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Pipet 10.0 ml of standard solution into a separate 100 ml volumetric flask, add about 50 ml 50/50 alcohol/water, 10.0 ml $Fe(NO_3)_3$ reagent and dilute to volume with 50/50 alcohol/water. Shake well. Filter the solutions until clear. Determine the absorbances of sample and standard in a 1 cm cell versus purified water in a suitable spectrophotometer at 525 nm.

Calculation:

$$\frac{Au \times Cs \times 10}{AS \times Sw} = \text{percent salicylic acid in finished ointment}$$

where:

Au = absorbance of unknown
As = absorbance of standard
Cs = standard weight in mg
Sw = sample weight in mg

4.7.3 pH. Disperse 20 grams of the ointment in 80 ml of purified water at 25° C, and determine the pH of the resulting solution potentiometrically using the USP method.

4.7.4 Loss on drying. Accurately weigh about 6.5 grams of ointment into a tared, aluminum drying dish. Place in an oven at 110° C ± 2° C for 2 hours; cool in a desiccator to constant weight, and weigh. Calculate the percent loss on drying.

4.7.5 Lather volume and permanence. Transfer 65 mg of ointment to a 100 ml glass stoppered, graduated cylinder, and add 20 ml of purified water at room temperature (25° ± 1° C). Shake vigorously for 30 seconds. The total volume shall be read immediately, and after a 30 minute interval.

4.7.6 Stability. Store the ointment in the original tubes for 10 weeks in an oven at 45° ± 4° C. After the 10 week period, remove the tubes from the oven and observe for gross separation by squeezing the contents from the tubes.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One box containing twelve (12) 4 oz tubes, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available ointment tube considered acceptable by the Food and Drug Administration (FDA) for the prioduct contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

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5.2.2.2 Unit package. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Six (6) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Six (6) units of issue shall be packed as specified in 5.4.3.5.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

* NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) Date of manufacture may appear on the unit package in lieu of the immediate container.

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5.4.2 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186, except that the date of manufacture shall be shown in lieu of date packed. Marking shall include the legends:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)"

and

"DO NOT PERMIT TO FREEZE".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-01-030-9067	SULFUR AND SALICYLIC ACID OINTMENT, 4 oz (113.4 grams), 12s

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6505-2017

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-37729 A SULFUR AND SALICYLIC ACID OINTMENT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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